

TITLE: WIRED AND WIRELESS NETWORK CONFIGURATION WITH RIP, VLANs, DHCP AND DNS

OBJECTIVES

- To understand and implement a combined wired and wireless network topology using Cisco Packet Tracer.
- To configure VLANs on a Layer 2 switch for network segmentation.
- To implement Inter-VLAN routing using Router-on-a-Stick method.
- To configure DHCP pools on routers to provide automatic IP address assignment.
- To set up a DNS server to resolve domain names to IP addresses.
- To configure RIPv2 as the dynamic routing protocol between two routers.
- To configure default routes and redistribute them via RIP.
- To capture and analyze RIP routing updates using Packet Tracer simulation mode.

THEORY

1. VLANs (Virtual Local Area Networks)

A VLAN is a logical grouping of network devices that are on different physical LAN segments but communicate as if they were on the same physical segment. VLANs are configured on a Layer 2 switch and allow network segmentation, improved security, and reduced broadcast domains. Devices in different VLANs cannot communicate without a Layer 3 device (router or Layer 3 switch). Inter-VLAN routing using a single router interface is known as Router-on-a-Stick, where subinterfaces are created for each VLAN with 802.1Q encapsulation.

2. DHCP (Dynamic Host Configuration Protocol)

DHCP is a network protocol that automatically assigns IP addresses and other network configuration parameters (subnet mask, default gateway, DNS server) to devices on a network. A DHCP server maintains a pool of IP addresses and leases them to clients upon request. Different DHCP pools can be configured for different subnets or VLANs, ensuring devices in each segment receive appropriate IP configuration automatically.

3. DNS (Domain Name System)

DNS is a hierarchical naming system that translates human-readable domain names (such as www.lab.com) into IP addresses that computers use to identify each other. A DNS server maintains records of domain name to IP address mappings. In a local network, a DNS server can be configured to resolve internal domain names, allowing users to access network resources by name rather than IP address.

4. RIP (Routing Information Protocol)

RIP is a distance-vector dynamic routing protocol that uses hop count as its routing metric. RIPv2 supports classless routing (CIDR) and uses multicast address 224.0.0.9 for routing updates transmitted over UDP port 520. Routers running RIP exchange their full routing tables every 30 seconds. The maximum hop count is 15, and a metric of 16 is considered unreachable. RIPv2 also supports the redistribution of static and default routes to neighboring routers using the default-information originate command.

5. Default Routing

A default route (0.0.0.0/0) serves as the gateway of last resort. It forwards packets to a specified next-hop when no specific route matches the destination in the routing table. The command used is: `ip route 0.0.0.0 0.0.0.0 [next-hop-address]`. Default routes can be redistributed into RIP so all routers in the network learn the path to unknown destinations.

NETWORK TOPOLOGY

The network topology consists of two Cisco 1941 routers connected via a Gigabit Ethernet crossover link. On the left side, Router0 connects to a Cisco 2950T-24 switch which links PC0, PC1, PC2, PC3, and Server0. On the right side, Router1 connects to a WRT300N wireless router (configured as an access point) which provides wireless connectivity to Laptop0, Laptop1, Smartphone0, and Smartphone1.

- Router0 (left) handles wired LAN with three VLANs and acts as the Inter-VLAN router and DHCP server for wired clients.
- Router1 (right) connects to the wireless segment and acts as DHCP server for wireless clients.
- The link between Router0 and Router1 uses the 10.0.0.0/30 subnet for point-to-point communication.
- WRT300N is configured as a pure wireless access point (connected via LAN port) to avoid NAT issues.

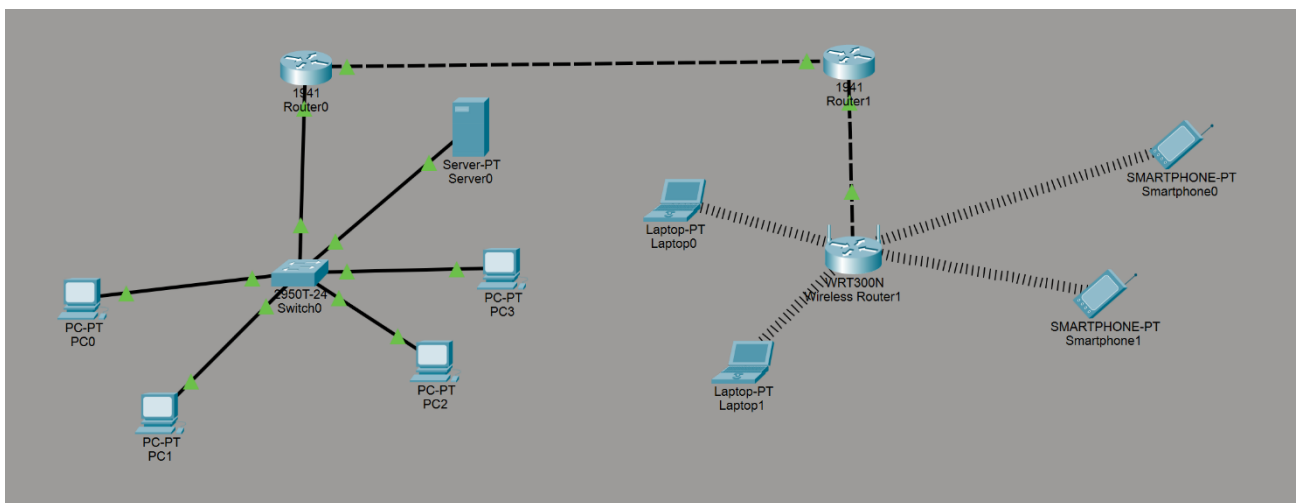


Fig: Network Topology

CONFIGURATION TABLE

Device IP Address Table:

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router0	Gi0/0.10	192.168.10.1	255.255.255.0	N/A
Router0	Gi0/0.20	192.168.20.1	255.255.255.0	N/A
Router0	Gi0/0.30	192.168.30.1	255.255.255.0	N/A
Router0	Gi0/1	10.0.0.1	255.255.255.252	N/A
Router1	Gi0/1	10.0.0.2	255.255.255.252	N/A
Router1	Gi0/0	192.168.40.1	255.255.255.0	N/A
Server0	Fa0	192.168.30.10	255.255.255.0	192.168.30.1
PC0	Fa0	DHCP (192.168.10.x)	255.255.255.0	192.168.10.1
PC1	Fa0	DHCP (192.168.10.x)	255.255.255.0	192.168.10.1
PC2	Fa0	DHCP (192.168.20.x)	255.255.255.0	192.168.20.1
PC3	Fa0	DHCP (192.168.20.x)	255.255.255.0	192.168.20.1
Laptop0/1	Wireless	DHCP (192.168.40.x)	255.255.255.0	192.168.40.1
Phone0/1	Wireless	DHCP (192.168.40.x)	255.255.255.0	192.168.40.1

VLAN Assignment Table:

VLAN ID	VLAN Name	Switch Ports	Devices
10	Group_A	Fa0/1, Fa0/2	PC0, PC1
20	Group_B	Fa0/4, Fa0/5	PC2, PC3
30	Server	Fa0/6	Server0
Trunk	All VLANs	Fa0/3	Router0 (Gi0/0)

DHCP Pool Configuration:

Pool Name	Network	Gateway	DNS Server	Clients
VLAN10_POOL	192.168.10.0/24	192.168.10.1	192.168.30.10	PC0, PC1
VLAN20_POOL	192.168.20.0/24	192.168.20.1	192.168.30.10	PC2, PC3
VLAN30_POOL	192.168.30.0/24	192.168.30.1	192.168.30.10	Server (Static)
WIRELESS	192.168.40.0/24	192.168.40.1	192.168.30.10	Laptops, Phones

Routing Configuration:

Device	Routing Type	Command / Network
Router0	RIPv2	network 192.168.10.0 / 192.168.20.0 / 192.168.30.0 / 10.0.0.0
Router0	Default Route	ip route 0.0.0.0 0.0.0.0 10.0.0.2
Router1	RIPv2	network 192.168.40.0 / 10.0.0.0
Router1	Default Route	ip route 0.0.0.0 0.0.0.0 10.0.0.1
Router1	Static Route	ip route 192.168.50.0 255.255.255.0 192.168.40.2 (removed after AP fix)

PROCEDURE

The following steps were performed to configure the network topology in Cisco Packet Tracer:

Step 1: Assign IP Addresses to Router0-Router1 Link

Both routers have only GigabitEthernet interfaces (no serial). The Gi0/1 interface on each router was used for the point-to-point WAN link.

On Router0:

```
Router0>enable
Router0#configure terminal
Router0(config)#interface gigabitEthernet 0/1
Router0(config-if)#ip address 10.0.0.1 255.255.255.252
Router0(config-if)#no shutdown
Router0(config-if)#exit
Router0(config)#end
Router0#write memory
```

On Router1:

```
Router1>enable
Router1#configure terminal
Router1(config)#interface gigabitEthernet 0/1
Router1(config-if)#ip address 10.0.0.2 255.255.255.252
Router1(config-if)#no shutdown
Router1(config-if)#exit
Router1(config)#end
Router1#write memory
```

Step 2: Assign IP to Router1 LAN Interface

Router1's Gi0/0 interface faces the WRT300N wireless router (connected via LAN port):

```
Router1#configure terminal
Router1(config)#interface gigabitEthernet 0/0
Router1(config-if)#ip address 192.168.40.1 255.255.255.0
Router1(config-if)#no shutdown
Router1(config-if)#exit
Router1(config)#end
Router1#write memory
```

Step 3: Configure WRT300N as Wireless Access Point

The cable from Router1 was connected to the LAN port (Ethernet 1) of WRT300N instead of the Internet port, to eliminate NAT. The WRT300N GUI was configured as follows:

Setting	Value
Connection to Router1	LAN Port (Ethernet 1) - No NAT
Router LAN IP	192.168.40.1 (same subnet as Router1)
Subnet Mask	255.255.255.0
DHCP Server	Disabled (Router1 handles DHCP)
SSID	LabWireless
Wireless Security	WPA2 Personal

Note: Since WRT300N is connected via LAN port, it acts as a transparent bridge/access point. Wireless devices communicate directly with Router1 without NAT.

Step 4: Configure VLANs on Switch0

VLANs were created and ports assigned based on physical connections verified using show interfaces status command:

```
Switch>enable
Switch#configure terminal
Switch(config)#vlan 10
Switch(config-vlan)#name Group_A
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Group_B
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name Server
Switch(config-vlan)#exit
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/6
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/3
```

```
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#end
Switch#write memory
```

Step 5: Configure Router0 Subinterfaces (Inter-VLAN Routing)

Router-on-a-Stick method was used. The physical interface Gi0/0 was brought up without an IP address, and subinterfaces were created for each VLAN:

```
Router0#configure terminal
Router0(config)#interface gigabitEthernet 0/0
Router0(config-if)#no ip address
Router0(config-if)#no shutdown
Router0(config-if)#exit
Router0(config)#interface gigabitEthernet 0/0.10
Router0(config-subif)#encapsulation dot1Q 10
Router0(config-subif)#ip address 192.168.10.1 255.255.255.0
Router0(config-subif)#no shutdown
Router0(config-subif)#exit
Router0(config)#interface gigabitEthernet 0/0.20
Router0(config-subif)#encapsulation dot1Q 20
Router0(config-subif)#ip address 192.168.20.1 255.255.255.0
Router0(config-subif)#no shutdown
Router0(config-subif)#exit
Router0(config)#interface gigabitEthernet 0/0.30
Router0(config-subif)#encapsulation dot1Q 30
Router0(config-subif)#ip address 192.168.30.1 255.255.255.0
Router0(config-subif)#no shutdown
Router0(config-subif)#exit
Router0(config)#end
Router0#write memory
```

Step 6: Configure DHCP Pools

On Router0 (for wired VLANs):

```
Router0#configure terminal
Router0(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.10
Router0(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.10
Router0(config)#ip dhcp excluded-address 192.168.30.1 192.168.30.10
Router0(config)#ip dhcp pool VLAN10_POOL
Router0(dhcp-config)#network 192.168.10.0 255.255.255.0
Router0(dhcp-config)#default-router 192.168.10.1
Router0(dhcp-config)#dns-server 192.168.30.10
Router0(dhcp-config)#exit
Router0(config)#ip dhcp pool VLAN20_POOL
Router0(dhcp-config)#network 192.168.20.0 255.255.255.0
Router0(dhcp-config)#default-router 192.168.20.1
Router0(dhcp-config)#dns-server 192.168.30.10
Router0(dhcp-config)#exit
Router0(config)#ip dhcp pool VLAN30_POOL
Router0(dhcp-config)#network 192.168.30.0 255.255.255.0
Router0(dhcp-config)#default-router 192.168.30.1
Router0(dhcp-config)#dns-server 192.168.30.10
Router0(dhcp-config)#exit
Router0(config)#end
```

```
Router0#write memory
```

On Router1 (for wireless clients):

```
Router1#configure terminal
Router1(config)#ip dhcp excluded-address 192.168.40.1 192.168.40.10
Router1(config)#ip dhcp pool WIRELESS
Router1(dhcp-config)#network 192.168.40.0 255.255.255.0
Router1(dhcp-config)#default-router 192.168.40.1
Router1(dhcp-config)#dns-server 192.168.30.10
Router1(dhcp-config)#exit
Router1(config)#end
Router1#write memory
```

Step 7: Configure DNS Server (Server0)

Server0 was assigned a static IP address and DNS service was enabled:

IP Configuration (Desktop > IP Configuration > Static):

Field	Value
IP Address	192.168.30.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.30.1
DNS Server	192.168.30.10

DNS Records added (Services > DNS > ON):

Name	Record Type	IP Address
www.lab.com	A Record	192.168.30.10
server.lab.com	A Record	192.168.30.10

Step 8: Configure Default Routes

Default routes were configured on both routers to ensure unknown traffic is forwarded to the neighboring router:

On Router0:

```
Router0#configure terminal
Router0(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.2
Router0(config)#end
Router0#write memory
```

On Router1:

```
Router1#configure terminal
Router1(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.1
Router1(config)#end
Router1#write memory
```

Step 9: Configure RIPv2 on Both Routers

On Router0:

```
Router0#configure terminal
Router0(config)#router rip
Router0(config-router)#version 2
```

```

Router0(config-router)#network 192.168.10.0
Router0(config-router)#network 192.168.20.0
Router0(config-router)#network 192.168.30.0
Router0(config-router)#network 10.0.0.0
Router0(config-router)#no auto-summary
Router0(config-router)#default-information originate
Router0(config-router)#exit
Router0(config)#end
Router0#write memory

```

On Router1:

```

Router1#configure terminal
Router1(config)#router rip
Router1(config-router)#version 2
Router1(config-router)#network 192.168.40.0
Router1(config-router)#network 10.0.0.0
Router1(config-router)#no auto-summary
Router1(config-router)#default-information originate
Router1(config-router)#exit
Router1(config)#end
Router1#write memory

```

RESULT

Routing Table Verification (Router0):

```

Router0#show ip route

C    10.0.0.0/30          via GigabitEthernet0/1   (WAN link)
C    192.168.10.0/24     via GigabitEthernet0/0.10 (VLAN 10)
C    192.168.20.0/24     via GigabitEthernet0/0.20 (VLAN 20)
C    192.168.30.0/24     via GigabitEthernet0/0.30 (VLAN 30)
R    192.168.40.0/24     [120/1] via 10.0.0.2   <- Learned via RIP from
Router1
S*   0.0.0.0/0           [1/0] via 10.0.0.2   <- Default route

```

DHCP Binding Verification (Router0):

```

Router0#show ip dhcp binding

IP address      Hardware address  Type
192.168.10.11   PC0 MAC address  Automatic
192.168.10.12   PC1 MAC address  Automatic
192.168.20.11   PC2 MAC address  Automatic
192.168.20.12   PC3 MAC address  Automatic

```

VLAN Verification (Switch0):

```

Switch#show vlan brief

VLAN  Name      Status  Ports
1     default  active  Fa0/7-Fa0/24, Gig0/1, Gig0/2
10    Group_A   active  Fa0/1, Fa0/2
20    Group_B   active  Fa0/4, Fa0/5
30    Server    active  Fa0/6

```


Connectivity Test Results:

Source	Destination	IP Address	Result
PC0 (VLAN 10)	PC2 (VLAN 20)	192.168.20.11	Success
PC0 (VLAN 10)	Server0 (VLAN 30)	192.168.30.10	Success
PC0 (VLAN 10)	Laptop0 (Wireless)	192.168.40.11	Success
Laptop0 (Wireless)	PC0 (VLAN 10)	192.168.10.11	Success
Laptop0 (Wireless)	Server0 (DNS)	192.168.30.10	Success
Router0	Router1	10.0.0.2	Success (80%)

RIP ROUTING UPDATE ANALYSIS

Capture Procedure:

1. Click the Simulation tab (bottom right, clock icon) in Packet Tracer.
2. Click Edit Filters, uncheck all protocols, and check only RIP.
3. Click Auto Capture / Play to start simulation.
4. Click any RIP envelope packet appearing on the topology between routers.
5. Select Inbound PDU Details or Outbound PDU Details to analyze the packet.

PDU Details Analysis:

Layer	Field	Value	Description
Layer 2 (Ethernet)	Destination MAC	0100.5E00.0009	RIPv2 Multicast MAC
Layer 2 (Ethernet)	Source MAC	Router0 MAC	Sending router interface
Layer 3 (IP)	Source IP	192.168.10.1	Router0 sending update
Layer 3 (IP)	Destination IP	224.0.0.9	RIPv2 multicast address
Layer 3 (IP)	TTL	255	Time to live
Layer 3 (IP)	Protocol	0x11 (UDP)	Transport protocol
Layer 4 (UDP)	Source Port	520	RIP standard port
Layer 4 (UDP)	Destination Port	520	RIP standard port
RIP	Command	2 (Response)	Router sending routing table
RIP	Version	2	RIPv2

Routes Advertised by Router0 to Router1:

Network	Metric (Hop Count)	Meaning
192.168.10.0/24	1	VLAN 10 - Group A (directly connected)
192.168.20.0/24	1	VLAN 20 - Group B (directly connected)
192.168.30.0/24	1	VLAN 30 - Server (directly connected)
10.0.0.0/30	1	WAN link (directly connected)

Routes Advertised by Router1 to Router0:

Network	Metric (Hop Count)	Meaning
192.168.40.0/24	1	Wireless segment (directly connected)
10.0.0.0/30	1	WAN link (directly connected)

RIP Protocol Summary:

Parameter	Value
Protocol	RIPv2
Transport	UDP Port 520
Update Interval	Every 30 seconds
Metric	Hop Count
Maximum Hops	15 (16 = unreachable)
Multicast Address	224.0.0.9
Command Type 1	Request (asking for routes)
Command Type 2	Response (sending routing table)
auto-summary	Disabled (no auto-summarization)
default-information originate	Redistributes default route into RIP

Using tracert from vlan 10 pc1(192.168.10.12) to smartphone connected to different network with IP 192.168.40.11 (currently at time of testing).

```
Cisco Packet Tracer PC Command Line 1.0
C:\>tracert 192.168.40.11

Tracing route to 192.168.40.11 over a maximum of 30 hops:

  1  0 ms      14 ms      0 ms      192.168.10.1
  2  0 ms      0 ms       0 ms      10.0.0.2
  3  *          10 ms      15 ms      192.168.40.11

Trace complete.
```

Fig: Trace route

DISCUSSION

In this experiment, a combined wired and wireless network was successfully configured using Cisco Packet Tracer. The wired segment used VLANs to segment the network into three logical groups: Group A (VLAN 10), Group B (VLAN 20), and Server (VLAN 30). Inter-VLAN routing was achieved using the Router-on-a-Stick technique on Router0, where subinterfaces with 802.1Q encapsulation were configured for each VLAN.

A significant challenge encountered during the lab was the NAT behavior of the WRT300N wireless router. When connected via its Internet port, WRT300N performed Network Address Translation (NAT), which prevented direct communication from wired PCs to wireless laptops. The traffic from wired to wireless was dropped because the WRT300N blocked unsolicited inbound connections through its NAT table. This was resolved by reconnecting the cable from Router1 to the LAN port of WRT300N, effectively using it as a transparent wireless access point without NAT. Router1 then directly managed DHCP for wireless clients.

RIPv2 was configured on both routers to enable dynamic route exchange. Router0 advertised the three VLAN networks and the WAN link, while Router1 advertised the wireless network. The default-information originate command was used to redistribute the default routes into RIP, allowing each router to learn the gateway of last resort. RIP updates were verified using both the show ip route command and Packet Tracer simulation mode, which showed RIPv2 packets being multicast to 224.0.0.9 on UDP port 520 every 30 seconds.

The DNS server was configured on Server0 with a static IP address, and DNS records were added for local domain names. All DHCP pools were verified using the show ip dhcp binding command, confirming that all wired PCs and wireless devices received correct IP addresses automatically.

CONCLUSION

The network topology was successfully implemented and all objectives were achieved. VLANs provided logical network segmentation on the wired switch, with Inter-VLAN routing allowing communication between different VLAN segments. Multiple DHCP pools ensured automatic IP address assignment for all devices. The DNS server resolved local domain names correctly. RIPv2 enabled dynamic route exchange between the two routers, and default routes were redistributed to provide gateway of last resort functionality. The NAT issue with WRT300N was resolved by using it as a pure wireless access point, enabling full bidirectional communication between wired and wireless segments. Overall, the experiment demonstrated the integration of multiple networking technologies including VLANs, DHCP, DNS, RIP, and wireless networking in a single cohesive network design.